PyCaret: Machine Learning Model Development Made Easy

Organisations use low code/no code (LC/NC) apps to construct new information systems swiftly in today's fast-paced digital world. This article introduces PyCaret a low code machine learning library written in Python.

PyCaret is a Python version of the Caret (short for Classification And REgression Training) package in the R programming language, and has many benefits.

Increased productivity: PyCaret, being a low code library, makes you more productive. With less time spent coding, you and your team can now focus on business problems.

Easy to use: This simple and easy to use machine learning library will help you to perform end-to-end ML experiments with fewer lines of code.

Business ready: PyCaret is a business-ready solution. It allows you to do prototyping quickly and efficiently from your choice of notebook environment.

You can create a virtual environment in Python and execute the following command to install the PyCaret complete version:

```
pip install pycaret [full]
```

A machine learning practitioner can do classification, regression, clustering, anomaly detection, natural language processing, association rules mining

```
from pycaret.datasets import get_data
dataset = get_data('iris')
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Figure 1: Loading the data set

```
from pycaret.classification import *
clf1 = setup(data =dataset, target = 'species')
```

Processing:

Initiated	. . . .	14:40:06
Status	. . . .	Preprocessing Data

Following data types have been inferred automatically, if they are correct press enter to continue or type 'quit' otherwise.

	Data Type
sepal_length	Numeric
sepal_width	Numeric
petal_length	Numeric
petal_width	Numeric
species	Label

Figure 2: PyCaret environment setup